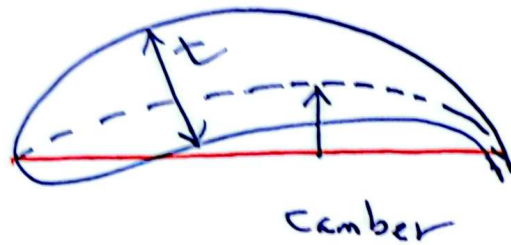




"predict lift from potential flow"

Potential Flow Over Airfoils



Airfoil nomenclature

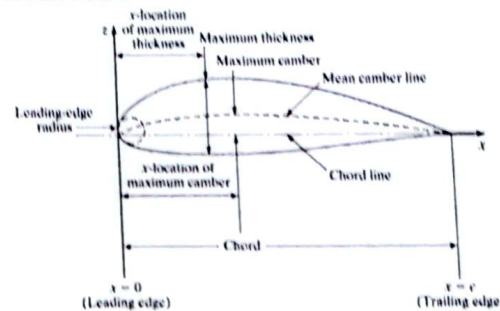
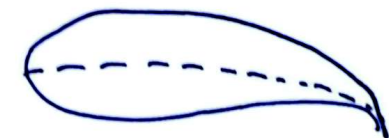
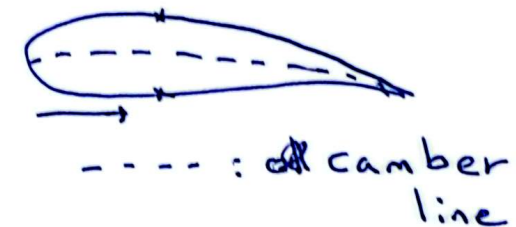
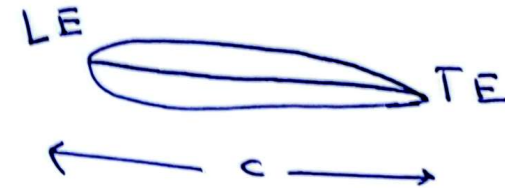


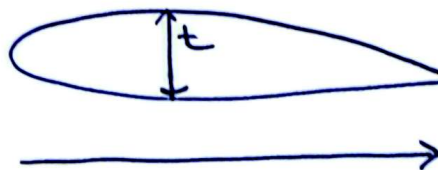
Figure 4.1: Airfoil nomenclature, Fig. 5.6 from AE-BC

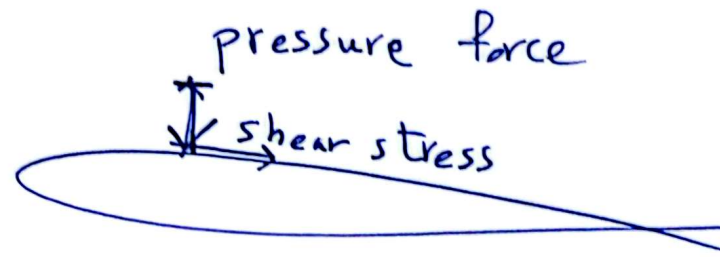
- ▶ **chord (c) / chord line** - Straight line connecting leading edge to trailing edge
- ▶ **mean camber line** - locus of points midway between upper and lower surface
- ▶ **thickness(x)** - upper surface - lower surface
- ▶ **camber** - location of maximum camber - for symmetric airfoil = 0

AE 319, EPG Pot. Airfoil, Basics, 132



camber is the same as chord for symm. airfoil





Forces on an airfoil

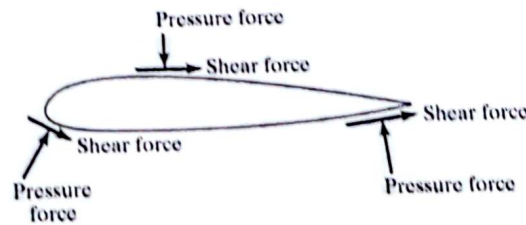


Figure 4.2: Forces on airfoil, Fig. 5.2 from AE-BC

- ▶ pressure $p(\vec{s})$ - this is the static pressure and always acts normal to the surface
- ▶ wall shear stress (force) or skin friction τ_w - always acts tangential to the surface

A.o.A:

Angle between
flow and
cord line

Forces on an airplane

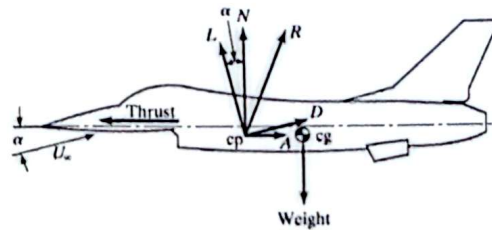
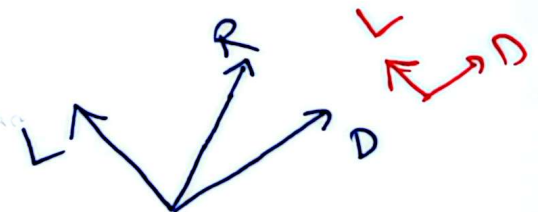
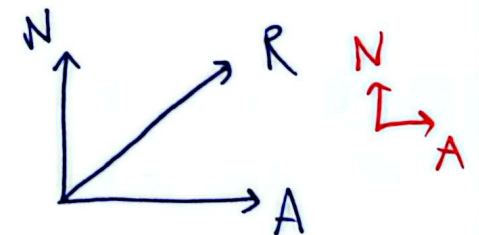
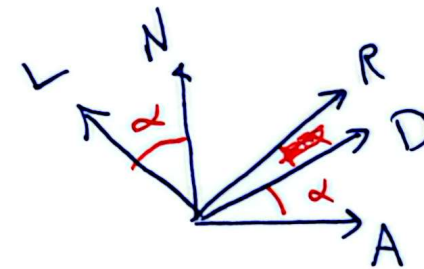
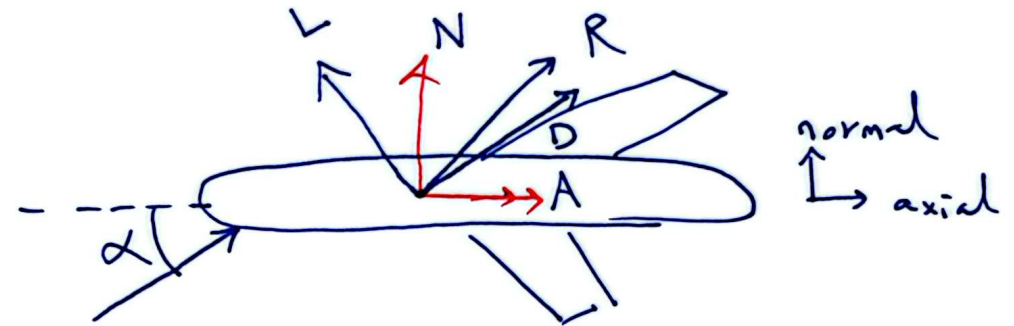


Figure 4.3: Forces on an airplane, Fig. 5.3 from AE-BC

- ▶ R = resultant force
- ▶ L = lift force - correspond force on an airfoil (2-D) l (per unit span)
- ▶ N = normal force
- ▶ A = axial force
- ▶ D = drag force - correspond force on an airfoil (2-D) d (per unit span)

AE 319, EPG Pot. Airfoil, Basics, 134



Forces on an airplane

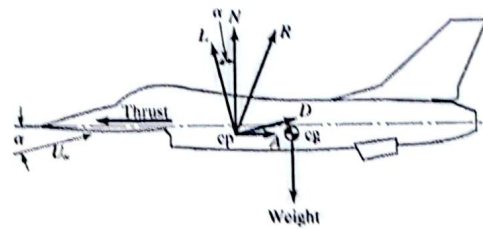


Figure 4.4: Forces on an airplane, Fig. 5.3 from AE-BC

$$\begin{bmatrix} D \\ L \end{bmatrix} = \begin{bmatrix} \cos \alpha & \sin \alpha \\ -\sin \alpha & \cos \alpha \end{bmatrix} \begin{bmatrix} A \\ N \end{bmatrix} \quad (4.1)$$

A and N
are easy to
measure

AE 319, EPG Pot. Airfoil, Basics, 135

$$D = A \cos \alpha + N \sin \alpha$$

$$L = -A \sin \alpha + N \cos \alpha$$

If α is small, $\begin{cases} D \approx A \\ L \approx N \end{cases}$

Moments on an airfoil

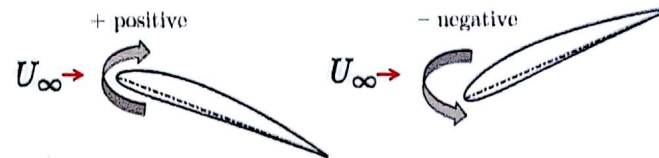


Figure 4.6: Sign convention for moments on an airfoil

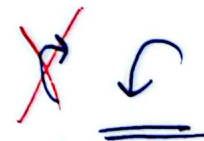
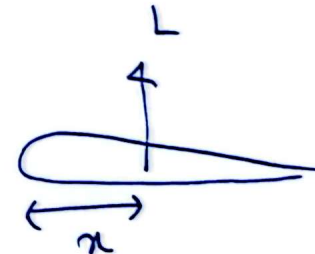
- Define around different points - usually leading edge m_0 or quarter chord $(c/4)$ $m_{c/4}$
- positive moment - pitch up, increases α
- negative moment - pitch down, decreases α

upward



→ good
for stability

$$M = L x$$



Non-Dimensional Numbers - wing

- Lift coefficient -

$$C_L = \frac{L}{q_\infty S} \quad (4.2)$$

- Drag coefficient -

$$C_D = \frac{D}{q_\infty S} \quad (4.3)$$

- Moment coefficient

$$C_M = \frac{M}{q_\infty S c} \quad (4.4)$$

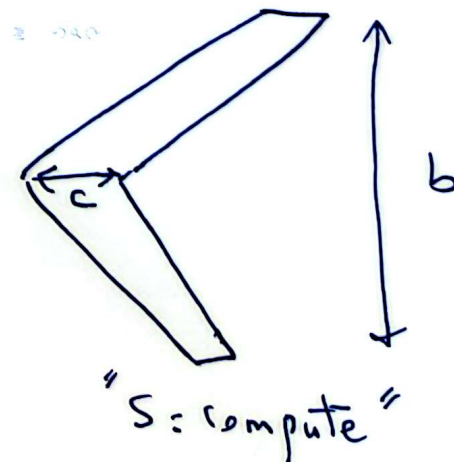
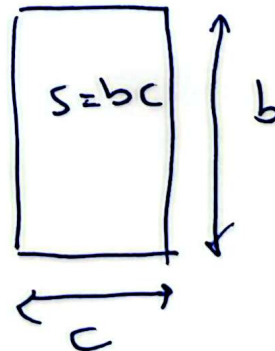
$$q_\infty = \frac{1}{2} \rho_\infty U_\infty^2 \rightarrow \text{dynamic pressure} \quad (4.5)$$

$$S = b \times c \rightarrow \text{planform area, } b \rightarrow \text{width of wing} \quad (4.6)$$

AE 319, EPG Pot. Airfoil, Basics, 138

For airfoils:

$$S = c \Rightarrow C_L = c_l$$



Airfoil Nomenclature - NACA 4 digit

NACA 2412

$$2 \rightarrow \text{max. camber in } \%c = 100m \quad (4.12)$$

$$4 \rightarrow \text{position of max. camber in } c/10 = (10p) \quad (4.13)$$

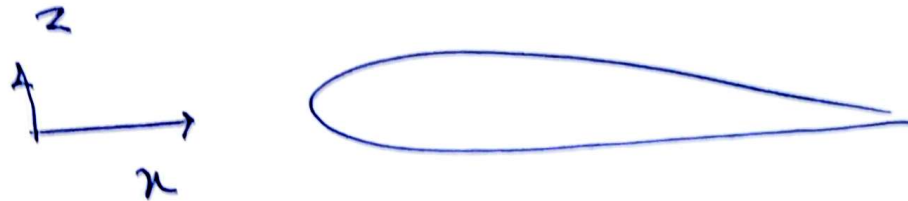
$$\underline{\underline{12}} \rightarrow \text{max. thickness in } \%c = 100t \quad (4.14)$$

Camber
formula

$$z_c = \begin{cases} \frac{x}{p^2} \left(2p - \frac{x}{c} \right), & 0 \leq x \leq pc \\ \frac{c-x}{(1-p)^2} \left(1 + \frac{x}{c} - 2p \right), & pc \leq x \leq c \end{cases} \quad (4.15)$$

thickness
formula

$$z_t = 5tc \left[0.2969 \sqrt{\frac{x}{c}} + (-0.1260) \left(\frac{x}{c} \right) + (-0.3516) \left(\frac{x}{c} \right)^2 + 0.2843 \left(\frac{x}{c} \right)^3 + (-0.1015) \left(\frac{x}{c} \right)^4 \right] \quad (4.16)$$



NACA 4 digit

$$\theta = \tan^{-1} \frac{dz_c}{dx} \quad (4.17)$$

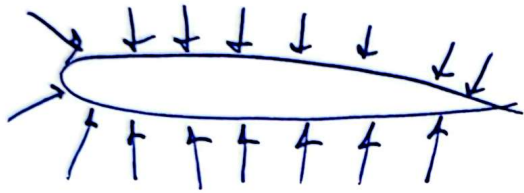
$$x_U = x - z_t \sin \theta, \quad z_U = z_c + z_t \cos \theta, \quad (4.18)$$

$$x_L = x + z_t \sin \theta, \quad z_L = z_c - z_t \cos \theta, \quad (4.19)$$

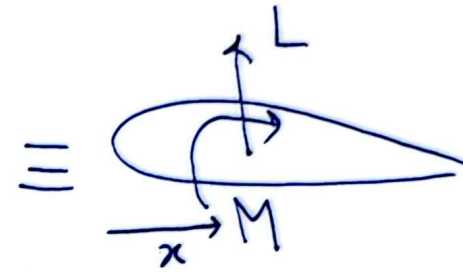
Equations are available in Abbott and Doenhoff "Theory of Wing Sections" or like me you can shamelessly copy it from Wikipedia - the ones on there are correct and the explanation is good.

https://en.wikipedia.org/wiki/NACA_airfoil

Refer to Abbott and Doenhoff "Theory of Wing Sections" for 5-digit and 6-digit sections.



distributed load



①

If you take moments with respect to center of pressure, moment is zero!

Center of Pressure

Center of pressure: Location of the single force - resultant R (per unit span) (see Fig. 4.10) that produces the same moment as the distributed load (from p and τ_w , see Fig. 4.2), on the airfoil

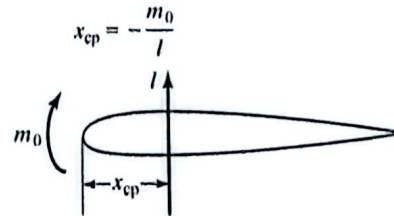
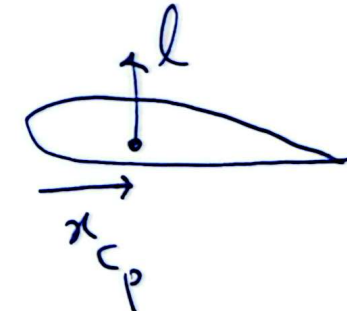


Figure 4.7: center of pressure, Fig 6.7 from AE-BC



②

$$m_0 = -l x_{cp} \quad (4.20)$$

$$x_{cp} = \frac{-m_0}{l} \quad (4.21)$$

Find M_{LE} in ① & ② :

① : $M_{LE} = M - Lx$

② : $M_{LE} = -L x_{cp} \Rightarrow x_{cp} = \frac{-M_{LE}}{L} = \frac{-m_0}{l}$

The negative point about center of pressure is that it changes with A.o.A (α)
 So, we define aerodynamic center, where:

Aerodynamic center

In general, the center of pressure for a cambered airfoil is

1. a function of angle of attack α
2. Note - $C_l = f(\alpha, \dots)$ - more on this soon.
3. Thus from a stability and control perspective x_{cp} is a difficult to use.

Aerodynamic center: x_{ac} The point at which the moment coefficient is invariant with Lift or C_l or C_L or α

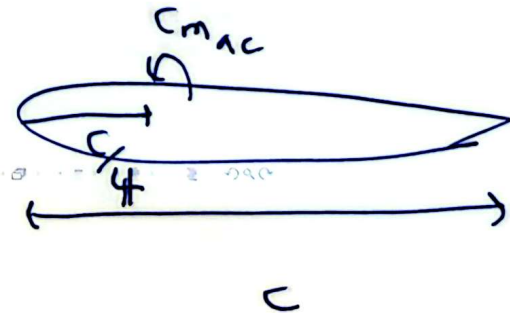
$$\text{For an airfoil at } x_{ac}: \frac{dC_{m,ac}}{dC_l} = \frac{dC_{m,ac}}{d\alpha} = 0 \quad (4.28)$$

For thin airfoils x_{ac} close to $c/4$

$$x_{ac} = \frac{dC_m}{dC_l} = 0$$

$$\text{or } \frac{dC_m}{d\alpha} = 0$$

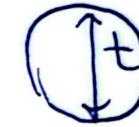
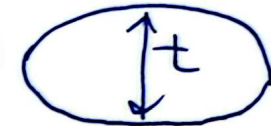
" C_m does not change with α "



remember

Source + Sink + uniform

Doublet + "



↳ creates bodies with thickness
w.o lift

Potential flow recap

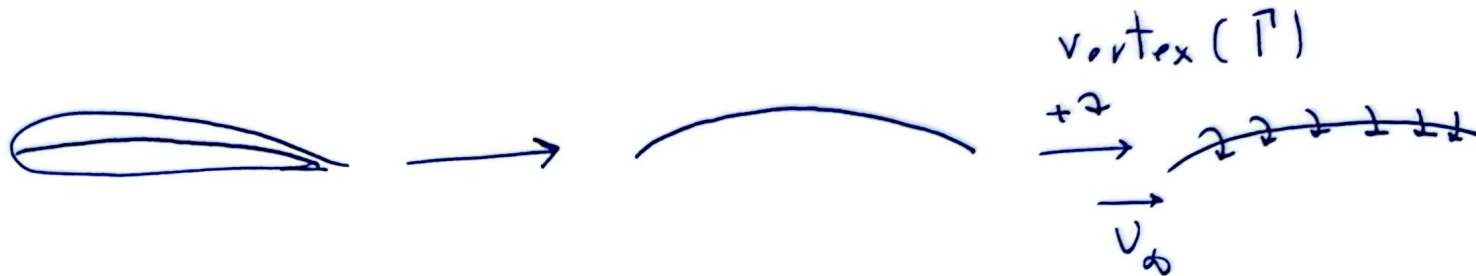
From what we learned about potential flow basics we can conclude a few things

- ▶ If we use only uniform flows, sources, sinks & doublets to generate flow around bodies we cannot generate lifting flows.
- ▶ But we can create flow over bodies with finite thickness.
- ▶ However, to get lift we need to add a vortex to the flow. Note - a vortex has infinite vorticity at its center. Everywhere but exactly at the vortex center, the flow is irrotational.
- ▶ To summarize uniform flow + source(s)/sink(s)/doublet(s) gives us thickness and we add vortex (or vortices) to get lift.

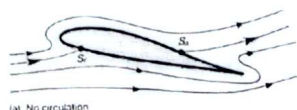
+ Vortex:

"gives lift"

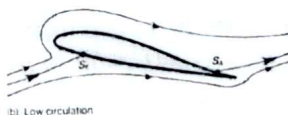
Let's for the moment assume that by using a combination of sources/sinks/doublets in a uniform flow we can generate non-lifting flow over an airfoil. Now, we add a vortex with strength Γ to this flow field. Based on the magnitude of Γ we choose we can get a few flow scenarios.



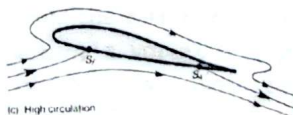
Kutta Condition



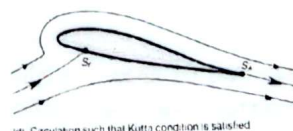
$$\Gamma_a = 0$$



$$0 = \Gamma_a < \Gamma_b < \Gamma_c$$

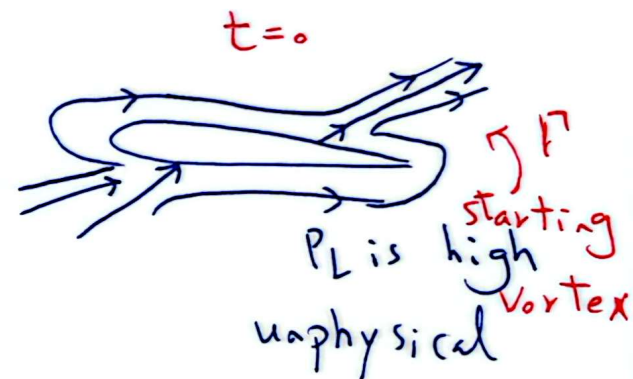


$$\Gamma_b < \Gamma_c < \Gamma_d$$

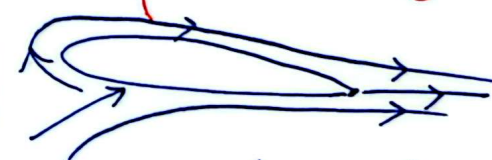


$\Gamma \rightarrow$ Kutta condition

Figure 4.11: Potential flow over an airfoil, adapted from AE-HCCV
AE 319, EPG Pot. Airfoil, Kutta, 149

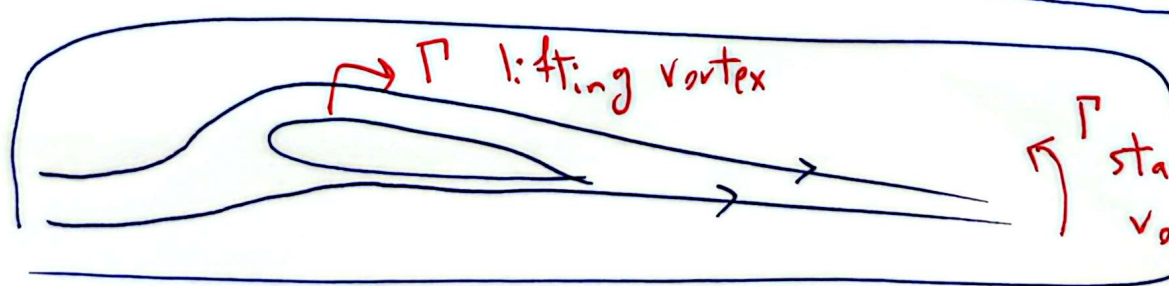


$t=t_{ss} \rightarrow \Gamma$: lifting vortex



physical

$t=t_{ss}$



Γ is such that
 $\Gamma_{TE} = 0$

$$\Gamma_{Lift} + \Gamma_{start} = 0$$

$$L = \rho_{\infty} V_{\infty} \Gamma$$

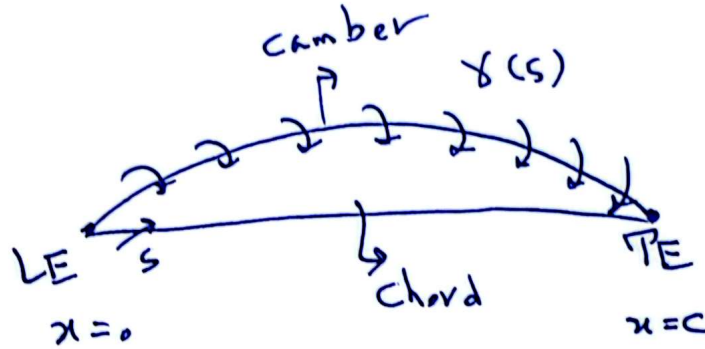
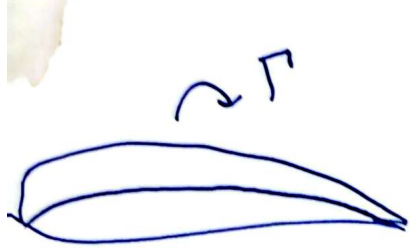
for airfoil

Kutta-Joukowski Theorem

From flow over a rotating cylinder, we derived lift per unit span to be,

$$l = \rho_{\infty} U_{\infty} \Gamma \quad (4.29)$$

This relationship is true for ALL potential flows. This can be easily verified.



$$\Gamma = \int_0^c \gamma(s) ds$$

$$\gamma(TE) = 0$$

$$\gamma(c) = 0$$

TE

Kutta condition for vortex sheet

So far we have considered point sources, sinks, doublets and vortices. Let us consider a distributed vorticity or a vortex sheet.

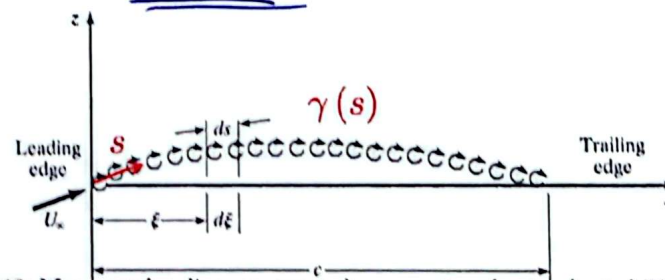


Figure 4.12: Mean camber line represented as a vortex sheet, adapted Fig 6.4 from AE-BC

Vortex strength per unit length is $\rightarrow \gamma(s)$ Note:

$$\Gamma = \int_0^c \gamma(s) ds \quad (\text{TEq. (6.1)}) \quad (4.30)$$

$$\text{Kutta condition} \rightarrow \gamma(c) = \gamma(TE) = 0 \quad (4.31)$$

Thin airfoil theory

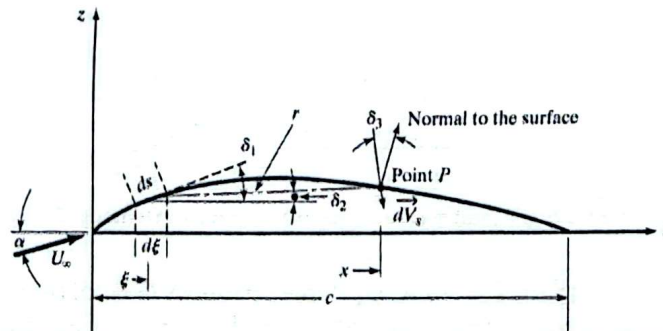


Figure 4.14: TAT geometry - no flow through camber line, Fig 6.5 from AE-BC
The elemental portion of the vortex sheet ds induces a velocity dV_s at point P . The normal component of this velocity is

$$dV_{s,n} = -\frac{\gamma ds \cos \delta_3}{2\pi r} \quad (4.32)$$

Camber line to
be streamline

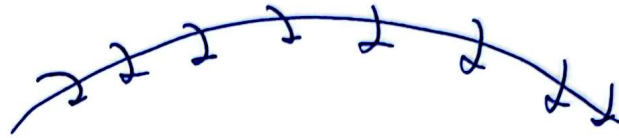
We have uniform flow
and vortices

For camber to be
streamline, $V_{s,n} = 0$

So normal velocity
from free stream
and vortices should
be zero!

$$V_{\text{vort},x} = \frac{\Gamma}{2\pi r}$$

$$d\Gamma = -\gamma ds \cos \delta_3$$



Thin airfoil theory

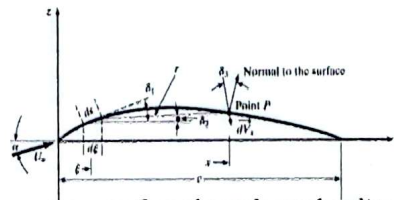


Figure 4.15: TAT geometry - no flow through camber line, Fig 6.5 from AE-BC

This is the contribution from the small piece ds . Therefore, the contribution from the entire sheet at this point P located arbitrarily at a distance x from the leading edge is

$$\boxed{V_{s,n}(x) = -\frac{1}{2\pi} \int_0^c \frac{\gamma(\xi) \cos \delta_2 \cos \delta_3}{(x-\xi) \cos \delta_1} d\xi} \quad (\text{TEq. (6.2)}) \quad (4.33)$$

Since x was chosen arbitrarily this is the expression for $V_{s,n}$ at any x i.e., point on the mean camber line.

Thin airfoil theory

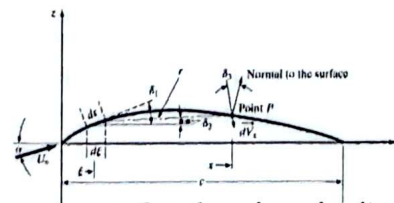


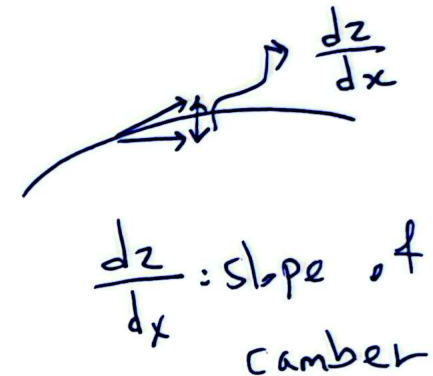
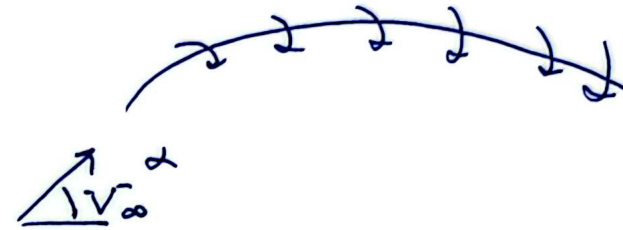
Figure 4.16: TAT geometry - no flow through camber line, Fig 6.5 from AE-BC

Similarly, the free-stream velocity U_∞ also contributes to the normal velocity at the P . Therefore, the normal component of the free-stream velocity is,

$$U_{\infty,n}(x) = U_\infty \sin(\alpha - \delta_p) \text{ \& } \delta_p = \tan^{-1} \frac{dz}{dx} \quad (4.34)$$

Here, dz/dx is the slope of the mean camber line. Finally,

$$U_{\infty,n}(x) = U_\infty \left(\alpha - \tan^{-1} \frac{dz}{dx} \right) \quad (\text{TEq. (6.3)}) \quad (4.35)$$



Fundamental Eqn. of TAT

Hence, the mathematical statement for no flow through the mean camber line becomes

$$\frac{1}{2\pi} \int_0^c \frac{\gamma(\xi) d\xi}{(x-\xi)} = U_{\infty} \left(\alpha - \frac{dz}{dx} \right) \quad (\text{TEq. (6.5)}) \quad (4.39)$$

Knowns: $z(x)$, U_{∞} & α

Unknown: $\gamma(\xi)$

This is the fundamental eq. of thin airfoil theory.

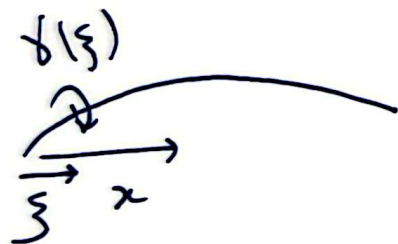
4.39 says that normal velocity component of free stream (right sid) = normal velocity " of vortices

AE 319, EPG Pot. Airfoil, TAT, 159

Note : z is known (camber formula), so $\frac{dz}{dx}$ is known, solve for $\gamma(\xi)$.

ξ : vortex location

x : velocity computation point



Fundamental Eqn. of TAT

unknown

known!

$$\begin{cases} V_{\infty} \\ \alpha \\ \frac{dz}{dx} \end{cases}$$

$$\frac{1}{2\pi} \int_0^c \frac{\gamma(\xi) d\xi}{(x-\xi)} = U_{\infty} \left(\alpha - \frac{dz}{dx} \right) \quad (\text{TEq. (6.5)}) \quad (4.40)$$

normal velocity of

This is an integral equation and not trivial to solve. Often it is easy to work in trigonometric functions. Hence, we carry out the following transformation.

Vortices

"normal velocity of free stream"

$$\text{Let, } \xi = \frac{c}{2} (1 - \cos \theta), \quad 0(\text{LE}) \leq \theta \leq \pi(\text{TE}) \quad (4.41)$$

$$x = \frac{c}{2} (1 - \cos \theta_0), \quad 0(\text{LE}) \leq \theta_0 \leq \pi(\text{TE}) \quad (4.42)$$

$$d\xi = \frac{c}{2} \sin \theta d\theta \quad (4.43)$$

$$\theta_0 = 0, \rightarrow x = 0(\text{LE}) \quad (4.44)$$

$$\theta_0 = \pi, \rightarrow x = c(\text{TE}) \quad (4.45)$$

Fundamental Eqn. of TAT

Finally, the fundamental eqn. of thin airfoil theory becomes,

$$\frac{1}{2\pi} \int_0^\pi \frac{\gamma(\theta) \sin \theta d\theta}{(\cos \theta - \cos \theta_0)} = U_\infty \left(\alpha - \frac{dz}{dx} \right) \quad (\text{TEq. (6.19)}) \quad (4.46)$$

Knowns: $z(x)$ (really $dz/dx(\theta)$), U_∞ & α

Unknown: $\gamma(\theta)$

General solution of fundamental eqn. of TAT

$$\frac{1}{2\pi} \int_0^\pi \frac{\gamma(\theta) \sin \theta}{(\cos \theta - \cos \theta_0)} d\theta = U_\infty \left(\alpha - \frac{dz_c}{dx} \right) \quad (4.47)$$

For a cambered airfoil $z(x) \neq 0$. We represent $dz(x)/dx$ as our old friend Fourier cosine series

$$\frac{dz}{dx} = (\alpha - A_0) + \sum_{n=1}^{\infty} A_n \cos(n\theta), \quad (\text{TEq. (6.22)}) \quad (4.48)$$

$$\text{here, } A_0 = \alpha - \frac{1}{\pi} \int_0^\pi \left(\frac{dz}{dx} \right) d\theta \quad (\text{TEq. (6.23)}) \quad (4.49)$$

$$A_n = \frac{2}{\pi} \int_0^\pi \left(\frac{dz}{dx} \right) \cos(n\theta) d\theta \quad (\text{TEq. (6.24)}) \quad (4.50)$$

After some math magic, the solution:

$$\gamma(\theta) = 2U_\infty \left[A_0 \frac{1 + \cos \theta}{\sin \theta} + \sum_{n=1}^{\infty} A_n \sin(n\theta) \right] \quad (\text{TEq. (6.20)}) \quad (4.51)$$

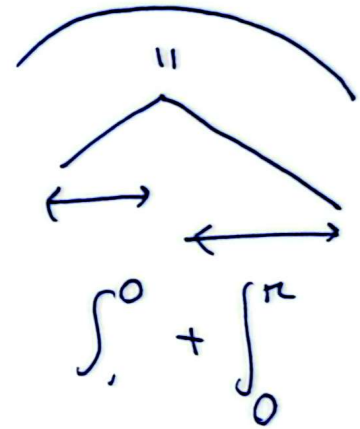
AE 319, EPG Pot. Airfoil, TAT Camb., 162

→ "crucial stage"

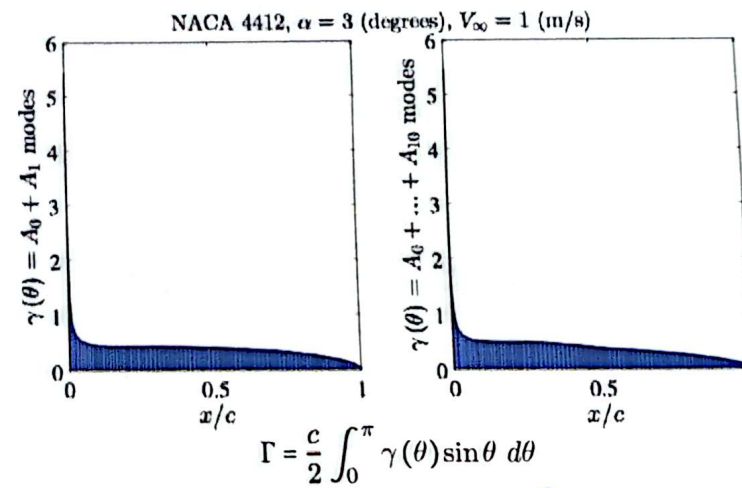
① Express $\frac{dz}{dx}$
in cosine
Fourier
Series

② Find A_0 & A_n

③ $\gamma(\theta)$ is a Sin Fourier series with the same A_0 & A_n as ①



Solution for NACA 2412



A_0 & A_1 are sufficient!

Thin airfoil theory - derived quantities

Lift coefficient:

$$l = \rho U_{\infty} \Gamma, C_l = \frac{l}{\frac{1}{2} \rho U_{\infty}^2 c}$$

$$\text{Therefore, } C_l = 2\pi \left(A_0 + \frac{A_1}{2} \right) \quad (4.56)$$

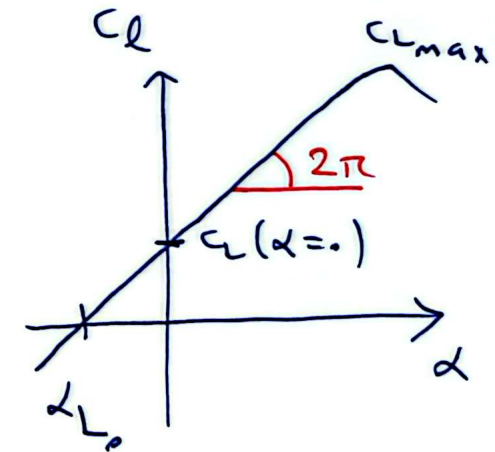
$$= 2\pi \left[\alpha + \frac{1}{\pi} \int_0^{\pi} \frac{dz}{dx} (\cos\theta - 1) d\theta \right] \quad (4.57)$$

$= -\alpha_{0l}, \text{ zero-lift angle of attack}$

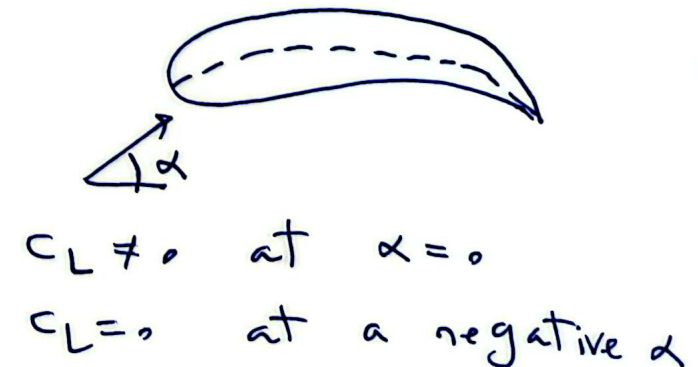
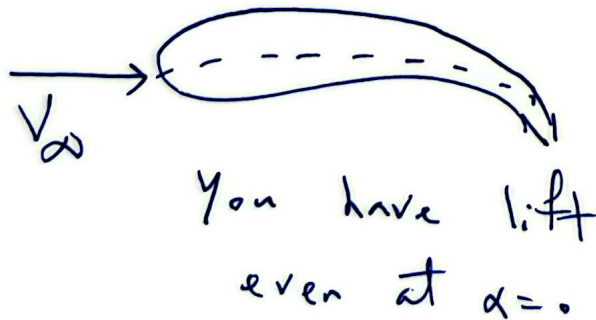
$$= 2\pi (\alpha - \alpha_{0l}) \quad (\text{TEq. (6.25)}) \quad (4.58)$$

$$\frac{dC_l}{d\alpha} = 2\pi \quad (4.59)$$

α_{L_0} depends
on camber
profile



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Thin airfoil theory - derived quantities

Lift coefficient - symmetric airfoil:

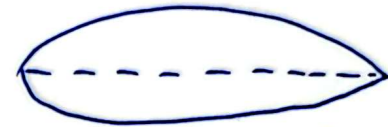
$$C_l = 2\pi \left[\alpha + \underbrace{\frac{1}{\pi} \int_0^\pi \frac{dz}{dx} (\cos\theta - 1) d\theta}_{=-\alpha_{0l}, \text{ zero-lift angle of attack}} \right] = 2\pi \left(A_0 + \frac{A_1}{2} \right)$$

In the case of a symmetric airfoil, $z = 0 \rightarrow dz/dx = 0$. Therefore, for a symmetric airfoil,

$$A_0 = \alpha \text{ \& } A_n = 0, n \geq 1 \quad (4.60)$$

$$\alpha_{0l} = 0 \quad (4.61)$$

$$C_l = 2\pi\alpha \quad (4.62)$$



Sym. airfoil:

$$\frac{dz}{dn} = 0 \Rightarrow \alpha_{0l} = 0$$

